HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Managed technology transfer initiatives for new product introductions, ensuring seamless scale-up from lab to commercial manufacturing and adherence to **cGMP compliance**.
- Collaborated with cross-functional teams including process chemists and engineers to drive project decisions and resolve technical challenges in a **fast-paced CDMO environment**.
- Interpreted complex **process data**, analytical results, and development reports to validate and qualify new manufacturing processes.
- Developed and executed **Python scripts** for data extraction and analysis, supporting the evaluation of process performance metrics for technology transfer projects.
- Supported the integration of new chemical processes into existing plant operations, ensuring alignment with **safety and environmental regulations**.
- Contributed to the design and implementation of **operational dashboards** to monitor key performance indicators during technology transfer campaigns.
- Containerized essential data processing tools using **Docker**, facilitating consistent deployment across development and pilot environments on **AWS EC2**.

SKILLS

Core Competencies – Leadership, Communication, Teamwork, Problem Solving, Project Management, Organizational Skills, Continuous Improvement

Technical Knowledge – Technology Transfer, API Pharmaceutical Manufacturing, Plant Operations, Chemical Process Engineering, Reaction Kinetics, Scale-Up, Crystallization, Isolation, Drying, Process Data Interpretation, Analytical Results Interpretation, Regulatory Guidelines, Safety Guidelines

Tools & Software – Python, SQL, Jira, Confluence, Splunk, Dynatrace, Docker, AWS (EC2, S3, Lambda, CloudWatch), Jenkins, Git, PostgreSQL, MySQL, Oracle, Power BI, Tableau, R/RStudio

Methodologies – Agile, Scrum, cGMP, Change Management

Domain Experience – Small Molecule API Manufacturing, Chemical Intermediates, HPAPI, Peptides, Pharmaceutical CDMO Operations

PROJECTS

API Process Scale-Up and Validation (HCL Technologies)

- Engineered a Python-based data processing pipeline to aggregate and analyze batch reaction data from pilot-scale runs, identifying key parameters for commercial scale-up.
- Developed automated reporting scripts in Python that summarized critical process deviations and quality control results, reducing manual reporting effort by **30%**.
- Integrated data from multiple sources using **SQL** across PostgreSQL and Oracle databases to build a unified view for process performance review.

Pharmaceutical Manufacturing Dashboard (MSc Applied Project)

- Designed and built interactive **Power BI** dashboards to visualize key process indicators (KPIs) for a simulated API manufacturing line, including yield, purity, and cycle time.
- Applied **DAX measures** and Power Query for complex data transformations and calculations, enabling stakeholders to track production efficiency and identify bottlenecks.
- Collaborated with simulated operations teams to gather requirements and ensure the dashboard effectively supported **decision-making** for process optimization.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)